

Price Discovery and Trading in Modern Prediction Markets

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A Rapidly Growing Market Spurs an Emerging Literature

- **Explosive growth**

- Monthly volume on Polymarket and Kalshi rose from under \$5 billion to \$24 billion, 9/2025–4/2026 (Pew Research 2026).

- **Trading patterns differ markedly from equity markets**

- Concentrated profits among informed traders (Akey et al., 2026; Gómez-Cram et al., 2026).
- Informed trading does not widen spreads (Bartlett and O'Hara, 2026).

- **Unsettled regulatory landscape**

- CFTC rulemaking roadmap and ongoing litigation over jurisdiction
- U.S. Commodity Futures Trading Commission (2026); Folkler (2026).

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We Focus on Price Discovery through Cross-Market Comparison

- Which platform discovers prices faster, and under what conditions?
 - A substantial subset of events is listed on both platforms.
 - We consider **matched markets** on the same underlying event, holding the fundamental fixed.
 - Explore how regulation, access, and clientele shape price discovery.

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 - Explore how regulation, access, and clientele shape price discovery.
- **We answer this in two complementary steps.**
 - **2024 U.S. presidential election** — one of the most liquid, closely watched events in prediction-market history.
 - **Full universe of matched markets** on Polymarket and Kalshi (Sep 13, 2023–Mar 29, 2026).

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■ 2024 Presidential Election

- Polymarket leads polls in predicting Trump's win.
- Persistent price gaps across four platforms; a violation of the law of one price.
- Polymarket leads Kalshi in price discovery.

■ Full Matched Sample of Polymarket and Kalshi Contracts

- Price discovery is bidirectional.
 - Polymarket leads politics and entertainment; Kalshi leads sports.
- Kalshi leads only when its relative volume or relative whale volume is high.
 - The lead vanishes when relative volume is low.
- Polymarket's lead is robust even when its relative volume is low.
- Polymarket's lead strengthens with insider risk

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Related Literature and Our Contribution

■ Prediction markets literature studied one platform at a time.

- Wolfers and Zitzewitz (2004); Arrow et al. (2008); Chernov, Elenev, and Song (2026); Akey et al. (2026); Gómez-Cram et al. (2026); Bartlett and O'Hara (2026)
- We provide the first study of cross-platform comparison of price formation.
 - Liquidity, large trades, category, and insider/manipulation risk matters.
 - Informs platform design and regulatory oversight.

■ Price discovery in fragmented markets

- Liquidity typically drives leadership, but mixed when naive traders dominate.
- Eun and Sabherwal (2003); Karolyi (2006); Chowdhry and Nanda (1991); Tetlock (2008)
- We find evidence consistent with a liquidity channel in this new market.
 - But regulatory frictions and insider/manipulation risk also matter.

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Conceptual Framework and Hypotheses

■ Three mechanisms motivate our hypotheses

- Information is incorporated through trading; deeper liquidity should speed price adjustment
 - Kyle (1985); Glosten and Milgrom (1985)
- Liquidity is self-reinforcing, creating a liquidity externality across venues
 - Pagano (1989); Collin-Dufresne and Fos (2016)
- Lighter oversight, pseudonymity, and open access raise incentives for insider trading and manipulation
 - Chowdhry and Nanda (1991)

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■ Hypotheses

- The more liquid venue attracts information acquisition; leads price discovery.
- Regulatory and access frictions generate deviations from the law of one price.
- In more loosely regulated venues, informed and insider trading contribute more, and large-trade order flow is especially informative.

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Presidential Election Analysis (Oct 4–Nov 5, 2024)

Key Differences Across Four Prediction Platforms:

Platform	Regulation	Identity / Access
Kalshi	CFTC-regulated exchange (DCM)	KYC-verified
Polymarket	Offshore, lightly regulated	Wallet-based, pseudonymous
PredictIt	CFTC no-action letter	Verified, position-capped
Robinhood	Access via CFTC-designated ForecastEx	Verified U.S. brokerage

- **Terminology:** A **platform** (e.g., Kalshi) lists **markets** tied to specific event outcomes (e.g., the DJT-Win market); traders exchange **contracts** within a market, each paying \$1 if the outcome occurs and \$0 otherwise.
- **Data:** obtained from official APIs and app screen recordings.
 - All four platforms: daily prices.
 - Kalshi and Polymarket: 30-minute returns.

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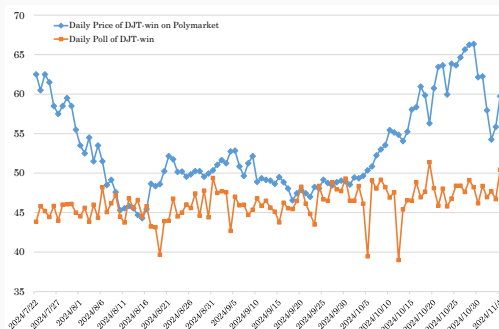
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Result 1: Polymarket Leads Polls in the 2024 Election



- Polymarket assigns Trump a higher probability than the polling average throughout the campaign.
- Polymarket returns lead polls, not the reverse.

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Result 2: Persistent Price Gaps Across Platforms

Average daily price difference across platforms, in cents (column minus row).

KH-Win above diagonal, **DJT-Win** below diagonal

	Kalshi	Polymarket	PredictIt	Robinhood
Kalshi	—	-3.27***	3.56***	2.39***
Polymarket	-2.59***	—	4.21***	6.09***
PredictIt	0.12	3.16***	—	-1.00
Robinhood	-2.16***	0.99**	-1.56	—

- **Average daily price difference across platforms economically sizable**
 - 4–5% of the DJT price and 15% of the KH price.
- **Price discrepancies can be calibrated using potential arbitrage profits.**
 - Kalshi versus Polymarket, matching opposing trades within a given interval.
 - 5-minute interval: \$2,581,976 before cost; \$1,008,230 after a 2% cost.
 - Hypothetical profits: geoblocking barred legal access to both legs, though offshore proxies could circumvent this.

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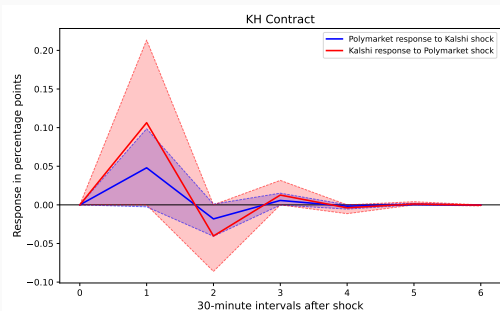
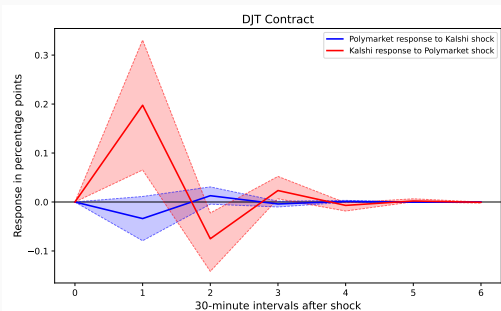
Result 3: Polymarket Leads Kalshi in Election Prediction

VAR regressions of 30-minute returns, separately for DJT-Win and KH-Win markets.

	DJT	KH
Polymarket $\rightarrow$ Kalshi	0.198***	0.106**
Kalshi $\rightarrow$ Polymarket	-0.034	0.048*

- **Polymarket leads Kalshi for both markets.**

- Reverse (Kalshi-to-Polymarket) coefficients insignificant for DJT and marginal for KH.



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Matched-Sample Analysis of the Full Universe of Polymarket and Kalshi Markets

- **Analyze the full universe of matched Polymarket-Kalshi markets.**
 - Sep 13, 2023–Mar 29, 2026
 - 1,972 matched market pairs in 1,533 matched events.
- **Matching procedure:**
 1. Classify events into eleven categories (e.g., Sports, Politics, Entertainment).
 2. Event matching via textual similarity + LLM-guided best-match selection.
 3. Within each matched event, match the most liquid markets.

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Matched example	Polymarket	Kalshi
Japan election	CRA wins 173+ seats	Centrist Reform Alliance wins above 172 seats
Trump on Joe Rogan	Appearance before election	Appearance before Nov 5, 2024

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Result 4 (Baseline): Price Discovery Is Bidirectional in the Full Sample

Panel VAR of 30-minute Polymarket and Kalshi returns.

	Equal weight	Volume weight
$P \rightarrow K$	0.100***	0.265***
$K \rightarrow P$	0.095***	0.636***

- Both directions are significant: price discovery is bidirectional.
 - Volume-weighted coefficients, which emphasize heavily traded markets, are larger: 0.265 (Polymarket $\rightarrow$ Kalshi) and 0.636 (Kalshi $\rightarrow$ Polymarket).

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 - Volume-weighted coefficients, which emphasize heavily traded markets, are larger: 0.265 (Polymarket $\rightarrow$ Kalshi) and 0.636 (Kalshi $\rightarrow$ Polymarket).
- **Next, we explore heterogeneity in leadership along four dimensions:**
 - Event category
 - Relative whale-trade activity
 - Relative trading volume
 - Insider/manipulation risk

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Result 5: Leadership Depends on Category

	Politics/elections	Entertainment	Sports
Volume-weighted lead-lag coefficients			
Polymarket → Kalshi	0.178***	0.059***	-0.074
Kalshi → Polymarket	-0.028	0.040	1.414***

- **Polymarket leads in politics and entertainment; Kalshi leads in sports.**
 - Tracks where trading concentrates: Polymarket captures most political volume, Kalshi most sports volume.
 - Entertainment is a partial exception: Kalshi has higher volume, yet Polymarket leads.
- **A natural question: what role does relative liquidity play?**

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Result 6: Kalshi's Lead Depends on Relative Volume; Polymarket's Does Not

We interact the panel VAR with relative Polymarket-to-Kalshi activity.

	Low P/K activity	Change in high P/K regime
Total trading activity		
Kalshi → Polymarket	0.220***	-0.211***
Polymarket → Kalshi	0.073***	-0.042
Whale trading activity		
Kalshi → Polymarket	0.196***	-0.186***
Polymarket → Kalshi	0.077***	-0.026

Whale trade: notional $\geq \max(\$1,000, \text{trailing 60-min 99th pctile})$, or $\geq \$500$ and $\geq 10\%$ of trailing 60-min market volume.

- Kalshi leads only when its relative volume or whale activity is high.
- Polymarket's lead is stable regardless of relative activity.
 - Lightly regulated design attracting informed traders regardless of liquidity.
 - E.g., Chowdhry and Nanda (1991); Hasbrouck (1995); Boehmer and Kelley (2009)

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Measuring Insider and Manipulation Risk for Results 7–8

For each market, we prompt an LLM with its title, resolution conditions, and resolution source, and ask it to score two risks from 1–10:

- ****Insider information:**** ability of a restricted group to know the decision, action, measurement, or decisive evidence materially before ordinary traders.
- ****Manipulation:**** ability of an identifiable actor or small group to cause, prevent, delay, or shape the qualifying real-world outcome or decisive resolution evidence. Price movement alone does not count.

Event	Insider	Manipulation
Harris/Democrat wins presidency	2	3
Ethereum all-time high	2	8
Best Director Oscar (Coogler)	7	3
@elonmusk profile picture change	9	10

Market-pair score is the average of the two platforms' individual-market scores.

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Result 7: Insider Risk Strengthens Polymarket's Lead, Weakens Kalshi's

Change from low to high insider-information risk (bottom to top risk quintiles)

	Politics/elections	Entertainment	Sports
Polymarket → Kalshi	0.111*** [4.69]	0.197*** [2.72]	-0.517 [-1.07]
Kalshi → Polymarket	-0.106*** [-2.75]	-0.340** [-2.18]	1.206 [1.63]

- **Insider risk strengthens Polymarket's lead in politics and entertainment.**

- Roughly doubles in politics;
- No lead in entertainment at low risk, but leads significantly at high risk.

- **Insider risk wipes out Kalshi's lead in the same two categories.**

- No significant shift in sports, for either direction.
- Caveat: risk scores may correlate with liquidity and participant composition.

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Result 8: Manipulation Risk Weakens Kalshi's Lead, Not Polymarket's

Change from low to high manipulation risk

	Politics/elections	Entertainment	Sports
Polymarket → Kalshi	-0.065 [-0.67]	0.070 [1.02]	-0.586 [-0.97]
Kalshi → Polymarket	-0.097* [-1.92]	-0.072 [-0.44]	0.108 [0.12]

- Polymarket's lead is unaffected by manipulation risk.
- Kalshi's lead wiped out in politics, marginally (10% level).

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- **We study price discovery across the major prediction-market platforms.**
 - Four platforms around the 2024 U.S. presidential election (Oct 4–Nov 5, 2024); the full matched universe of Polymarket-Kalshi market pairs (Sep 2023–Mar 2026).
- **Price discovery is not concentrated on one platform.**
 - Persistent price gaps and a Polymarket lead in the election sample;
 - Bidirectional leadership on average in the full sample;
 - Varying with category, relative volume, whale activity, and insider/manipulation risk.
- **Policy tension:** the openness and liquidity that speed price discovery also make insider trading and manipulation easier.

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